Topic: DFA

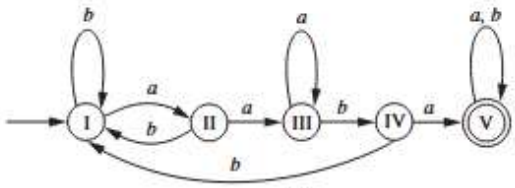
Question:

In each part below, draw an FA accepting the indicated language over {a, b}.

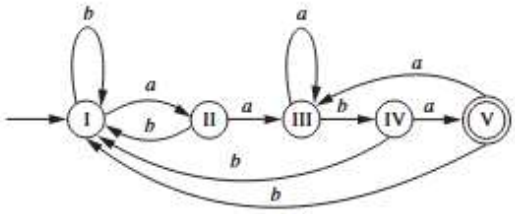
a. The language of all strings containing exactly two a's.
b. The language of all strings containing at least two a's.
c. The language of all strings that do not end with ab.
d. The language of all strings that begin or end with aa or bb.
e. The language of all strings not containing the substring aa.
f. The language of all strings in which the number of a's is even.
g. The language of all strings in which both the number of a's and the number of b's are even.
h. The language of all strings containing no more than one occurrence of the string aa. (The string aaa contains two occurrences of aa.)
i. The language of all strings in which every a (if there are any) is followed immediately by bb.
j. The language of all strings containing both bb and aba as substrings.
k. The language of all strings containing both aba and bab as substrings.

Question:

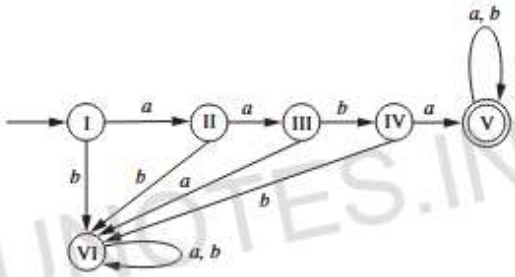
For each of the FAs pictured in below figure, give a simple verbal description of the language it accepts.



(a)



(b)



(c)

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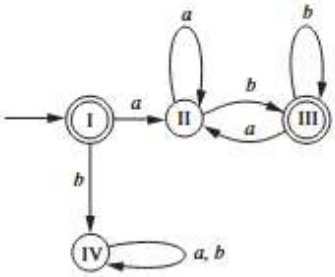

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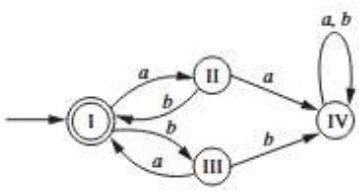


(e)

Question:

a. Draw a transition diagram for an FA that accepts the string abaa and no other strings.
b. For a string x ∈ {a, b}∗ with |x| = n, how many states are required for an FA accepting x and no other strings? For each of these states, describe the strings that cause the FA to be in that state.
c. For a string x ∈ {a, b}∗ with |x| = n, how many states are required for an FA accepting the language of all strings in {a, b}∗ that begin with x? For each of these states, describe the strings that cause the FA to be in that state.

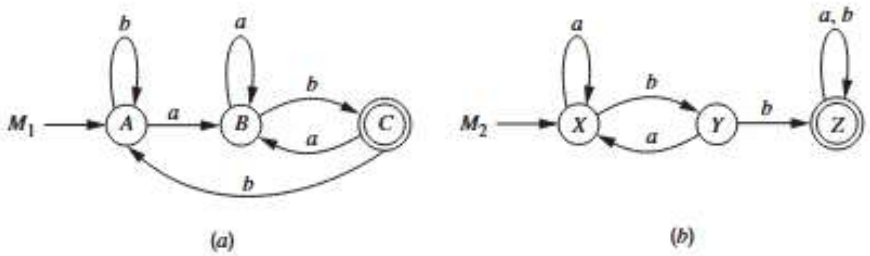
Question:

a. Let L3 is a language with the set of strings in {0, 1}∗ that are binary representations of integers divisible by 3. Construct FA and write the transition table for the same.
b. Draw a transition diagram for an FA accepting L5 .

Question:

Let M1 and M2 be the FAs pictured in below figure, accepting languages L1 and L2 , respectively. Draw FAs accepting the following languages.

a. L1 ∪ L2
b. L1 ∩ L2
c. L1 − L2



(a) (b)

Question:

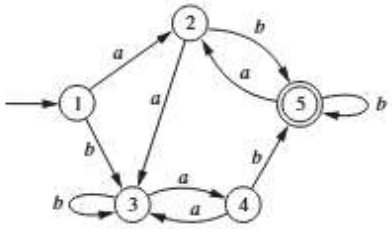
For each of the following languages, draw an FA accepting it.

a. {a, b}∗ {a}
b. {bb, ba}∗
c. {a, b}∗ {b, aa}{a, b}∗
d. {bbb, baa}∗ {a}
e. {a} ∪ {b}{a}∗ ∪ {a}{b}∗ {a}
f. {a, b}∗ {ab, bba}
g. {b, bba}∗ {a}
h. {aba, aa}∗ {ba}

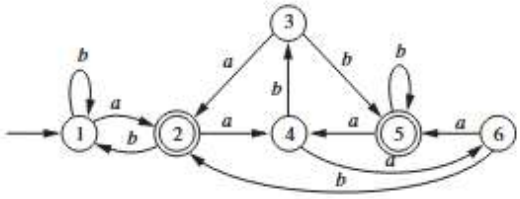
(Hint: represent the language L in terms of regular expression i.e., L(r), and then draw FA for the regular expression)

Question:

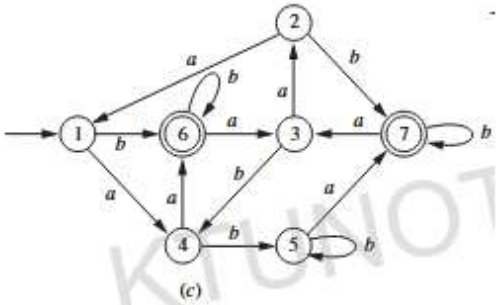
For each of the FAs pictured in below figures, use the minimization algorithm to find a minimum-state FA recognizing the same language. (It is possible that the given FA may already be minimal.)



(a)



(b)



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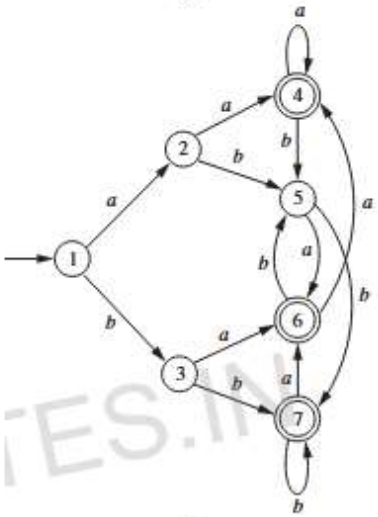

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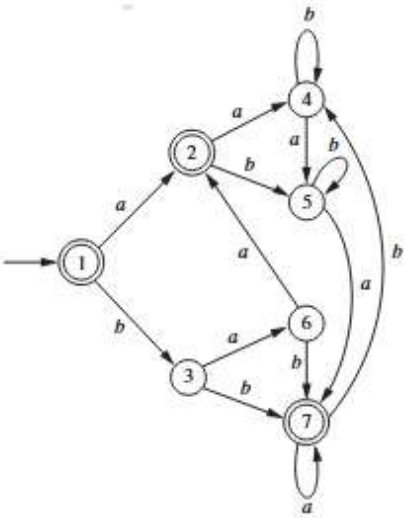
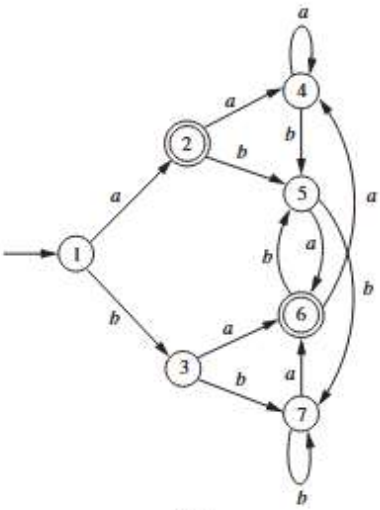
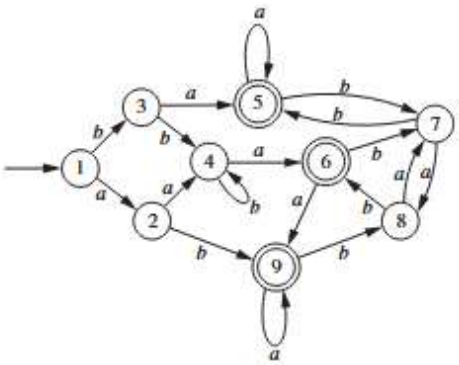


(e)



(f)



(g)

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